jashwanthuser@localhost:~ — vim bank.c

```c
#include<stdio.h>
int main(){
    int n,r,i,j,k;
    printf("Enter number of processes:");
    scanf("%d",&n);
    printf("Enter number of resource types:");
    scanf("%d",&r);
    int alloc[n][r],max[n][r],need[n][r],avail[r];
    int finish[n],safe_sequence[n];
    int count=0,safeFound;
    printf("Enter aloocation matrix:\n");
    for(i=0;i<n;i++){
        for(j=0;j<n;j++){
            scanf("%d",&alloc[i][j]);
        }
    }
    printf("Enter max matrix:\n");
    for(i=0;i<n;i++){
        for(j=0;j<r;j++){
            scanf("%d",&max[i][j]);
        }
    }
    printf("Enter available resources:\n");
    for(i=0;i<r;i++){
        scanf("%d",&avail[i]);
    }
    for(i=0;i<n;i++){
```

4,1-4 Top

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```
printf("Enter max matrix:\n");
for(i=0;i<n;i++){
    for(j=0;i<r;j++){
        scanf("%d",&max[i][j]);
    }
}
printf("Enter available resources:\n");
for(i=0;i<r;i++){
    scanf("%d",&avail[i]);
}
for(i=0;i<n;i++){
    finish[i]=0;
}
for(i=0;i<n;i++){
    for(i=0;i<r;i++){
        need[i][j]=max[i[j] - alloc[i][j];
    }
}
while(count<n){
    safeFound=0;
    for(i=0;i<n;i++){
        if(finish[i]==0){
           int safe1;
           for(j=0;j<r; j++){
               if(need[i][j]>avail[j]);
                                                    29,1-4        34%
```

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```
                safe =0;
                break;
            }
        }
        if(safe){
            for(k=0;k<r;k++){
                avail[k]=avail[k] + alloc[i][k];
            }
            safe_sequence[count++]=1;
            finish[i]=1;
            safeFound=1;
            count++;
        }
    }
    if(!safeFound){
        printf("System is in unsafe state! May occur the deadlock\n");
        return 0;
    }
}
printf("Safe sequence is:");
for(i=0;i<n;i++){
    printf("P%d",safe_squence[i]);

return 0;
```

66,1-4 Bot

jashwanthuser@localhost:~

```
Enter number of processes:3
Enter number of resource types:3
Enter aloocation matrix:
4
5
6
1
2
2
2
Enter max matrix:
7
5
3
2
1
2
1
Enter availaable resources:
5 4 6

Safe sequence is:P0 P1 P2

[jashwanthuser@localhost ~]$ vim bank.c
[jashwanthuser@localhost ~]$
```